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1 /Users/mehakagrawal/PycharmProjects/
  Final_Dissertation/.venv/bin/python /Users/
  mehakagrawal/Desktop/Final_Dissertation/Codes/4.
  tert_3month_de_analysis/tert_3month_de_analysis.py
2 STEP 1: Building combined count matrix from 13
  featureCounts files
3 Found 13 featureCounts files
4   M3_WT_234: 36.1M reads mapped
5   M3_Het_235: 14.9M reads mapped
6   M3_Het_251: 36.4M reads mapped
7   M3_WT_253: 37.0M reads mapped
8   M3_MUT_254: 29.1M reads mapped
9   M3_WT_255: 34.9M reads mapped
10  M3_MUT_256: 43.7M reads mapped
11  M3_WT_257: 41.1M reads mapped
12  M3_Het_258: 27.7M reads mapped
13  M3_Het_264: 47.5M reads mapped
14  M3_WT_270: 10.6M reads mapped
15  M3_Het_271: 36.7M reads mapped
16  M3_MUT_273: 44.1M reads mapped
17
18 Combined count matrix: 32105 genes x 13 samples
19 Saved: tert3m_raw_counts.csv
20
21
22 STEP 2: Metadata
23           animal_id genotype  age
24 M3_WT_234         234      WT   3mo
25 M3_Het_235         235     Het   3mo
26 M3_Het_251         251     Het   3mo
27 M3_WT_253         253      WT   3mo
28 M3_MUT_254         254     MUT   3mo
29 M3_WT_255         255      WT   3mo
30 M3_MUT_256         256     MUT   3mo
31 M3_WT_257         257      WT   3mo
32 M3_Het_258         258     Het   3mo
33 M3_Het_264         264     Het   3mo
34 M3_WT_270         270      WT   3mo
35 M3_Het_271         271     Het   3mo
36 M3_MUT_273         273     MUT   3mo
37

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38 Groups: {'WT': 5, 'Het': 5, 'MUT': 3}
39
40
41 STEP 3: Pre-filtering
42 Before: 32105 genes
43 After (>= 10 reads in >= 3 samples): 18522 genes
44 Removed: 13583
45
46
47 STEP 4: Normalisation + PCA
48 PCA variance: PC1=53.2%, PC2=8.5%, PC3=7.8%
49
50
51 DE Analysis: A: MUT vs WT (3mo)
52 Using None as control genes, passed at DeseqDataSet
  initialization
53 Fitting size factors...
54 ... done in 0.00 seconds.
55
56 Fitting dispersions...
57 ... done in 1.77 seconds.
58
59 Fitting dispersion trend curve...
60 ... done in 0.20 seconds.
61
62 Fitting MAP dispersions...
63 ... done in 2.27 seconds.
64
65 Fitting LFCs...
66 ... done in 1.44 seconds.
67
68 Calculating cook's distance...
69 ... done in 0.01 seconds.
70
71 Replacing 0 outlier genes.
72
73 Running Wald tests...
74 ... done in 0.44 seconds.
75
76 Log2 fold change & Wald test p-value: genotype MUT
  vs WT
```

			baseMean	log2FoldChange
	...	pvalue	padj	
78	Geneid			...
79	ENSDARG000000104632	27.385337		-0.397447
	...	0.445041	0.999855	
80	ENSDARG000000100660	280.199307		0.219747
	...	0.599506	0.999855	
81	ENSDARG000000100422	45.063546		1.155630
	...	NaN	NaN	
82	ENSDARG000000102128	128.521303		0.085558
	...	0.868399	0.999855	
83	ENSDARG000000103095	274.560056		0.319052
	...	0.251135	0.999855	
84	...		...	...
	.	...	...	
85	ENSDARG000000101185	50.864420		1.275981
	...	0.000045	0.110347	
86	ENSDARG000000102248	8.897637		-0.191274
	...	0.784828	0.999855	
87	ENSDARG000000102790	77.325523		-0.340791
	...	0.507792	0.999855	
88	ENSDARG000000100589	20.453289		0.467207
	...	0.485225	0.999855	
89	ENSDARG000000099551	5.872410		-0.536178
	...	0.501458	0.999855	
90				
91	[18522 rows x 6 columns]			
92				
93	Significant DEGs (q<0.05, log2FC >1.0): 0			[pi0=1.000]
94	Up: 0			
95	Down: 0			
96				
97	Top 15 by q-value:			
98				
99				
100	DE Analysis: B: Het vs WT (3mo)			
101	Using None as control genes, passed at			
	DeseqDataSet initialization			

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102 Fitting size factors...
103 ... done in 0.01 seconds.
104
105 Fitting dispersions...
106 ... done in 1.68 seconds.
107
108 Fitting dispersion trend curve...
109 ... done in 0.21 seconds.
110
111 Fitting MAP dispersions...
112 ... done in 1.87 seconds.
113
114 Fitting LFCs...
115 ... done in 1.18 seconds.
116
117 Calculating cook's distance...
118 ... done in 0.01 seconds.
119
120 Replacing 0 outlier genes.
121
122 Running Wald tests...
123 ... done in 0.44 seconds.
124
125 Log2 fold change & Wald test p-value: genotype Het
    vs WT
126
127 Geneid      baseMean  log2FoldChange
      ...      pvalue      padj
      ...
128 ENSDARG00000104632  28.352465      0.108583
      ... 0.817364 0.999554
129 ENSDARG00000100660  234.142437     -0.059943
      ... 0.866556 0.999554
130 ENSDARG00000100422   33.562607      0.517677
      ... 0.194404 0.999554
131 ENSDARG00000102128  111.447328     -0.059558
      ... 0.897893 0.999554
132 ENSDARG00000103095  241.631568      0.165133
      ... 0.467310 0.999554
133 ...
      ...
      ... ..

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133 .      ...      ...
134 ENSDARG000000101185    50.605337      1.252285
      ... 0.010713 0.838098
135 ENSDARG000000102248    7.406850      -0.415049
      ... 0.466543 0.999554
136 ENSDARG000000102790    85.138896      0.307937
      ... 0.501776 0.999554
137 ENSDARG000000100589    20.005788      0.546829
      ... 0.345844 0.999554
138 ENSDARG000000099551    7.372319      0.496200
      ... 0.379861 0.999554
139
140 [18522 rows x 6 columns]
141
142 Significant DEGs (q<0.05, |log2FC|>1.0): 1    [pi0=
      1.000]
143   Up: 1
144   Down: 0
145
146 Top 15 by q-value:
147   ENSDARG000000100265      log2FC=+4.43  qvalue=
      1.01e-02
148
149
150 DE Analysis: C: MUT vs Het (3mo)
151 Using None as control genes, passed at
      DeseqDataSet initialization
152 Fitting size factors...
153 ... done in 0.00 seconds.
154
155 Fitting dispersions...
156 ... done in 1.49 seconds.
157
158 Fitting dispersion trend curve...
159 ... done in 0.19 seconds.
160
161 Fitting MAP dispersions...
162 ... done in 2.10 seconds.
163
164 Fitting LFCs...
165 ... done in 1.25 seconds.

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166
167 Calculating cook's distance...
168 ... done in 0.01 seconds.
169
170 Replacing 0 outlier genes.
171
172 Running Wald tests...
173 ... done in 0.52 seconds.
174
175 Log2 fold change & Wald test p-value: genotype MUT
    vs Het
176
177 Geneid      baseMean  log2FoldChange
      ...      padj
178 ENSDARG00000104632  32.792484  -0.519580
      ... 0.324605 0.999873
179 ENSDARG00000100660  306.605075  0.260979
      ... 0.608838 0.999873
180 ENSDARG00000100422  58.589045  0.650598
      ...      NaN      NaN
181 ENSDARG00000102128  140.807515  0.131192
      ... 0.784478 0.999873
182 ENSDARG00000103095  330.862168  0.135403
      ... 0.571798 0.999873
183 ...
      ...
184 ENSDARG00000101185  88.958632  0.007795
      ... 0.989797 0.999873
185 ENSDARG00000102248  8.188144  0.218954
      ... 0.713514 0.999873
186 ENSDARG00000102790  101.616022  -0.668832
      ... 0.150353 0.999873
187 ENSDARG00000100589  29.298148  -0.103236
      ... 0.849233 0.999873
188 ENSDARG00000099551  8.826369  -1.070213
      ... 0.090926 0.999873
189
190 [18522 rows x 6 columns]
191

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192 Significant DEGs (q<0.05, |log2FC|>1.0): 1 [pi0=
    1.000]
193   Up: 0
194   Down: 1
195
196 Top 15 by q-value:
197   ENSDARG00000099419          log2FC=-5.46   qvalue=
    4.53e-04
198
199
200 STEP 6: Volcano plots
201   Saved: volcano_MUT_vs_WT.png
202   Saved: volcano_Het_vs_WT.png
203   Saved: volcano_MUT_vs_Het.png
204
205
206 STEP 7: Cross-dataset overlap (3mo vs 18mo
    genotype effect)
207 Genes in 3mo dataset: 18522
208 Genes in 18mo dataset: 22253
209 Shared gene IDs: 16873
210 3mo-only genes: 1649
211 18mo-only genes: 5380
212
213 3mo MUT vs WT DEGs: 0
214 18mo MUT vs WT DEGs (by qvalue): 5541
215 DEG overlap: 0
216
217
218 SUMMARY
219
220 Dataset: 13 muscle samples, 3 months old zebrafish
221   WT (tert +/+): n=5
222   Het (tert +/-): n=5
223   MUT (tert -/-): n=3
224
225 Genes analysed: 18522
226
227 Comparison A - MUT vs WT:
228   DEGs: 0 (up:0 down:0)
229
```

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230 Comparison B - Het vs WT:
231   DEGs: 1 (up:1 down:0)
232
233 Comparison C - MUT vs Het:
234   DEGs: 1 (up:0 down:1)
235
236 PCA: PC1=53.2%, PC2=8.5%
237
238 All outputs saved to: /Users/mehakagrawal/Desktop/
    Final_Dissertation/Outputs/4.
    tert_3month_de_analysis
239
240 Process finished successfully!
241
242 Process finished with exit code 0
243
```